Recap

Multi Modal RAG → data ingestion → Text

↓ Retrieval + Generation

LLM

RAG

different sources of input and output

Text, images, audio, video

speech

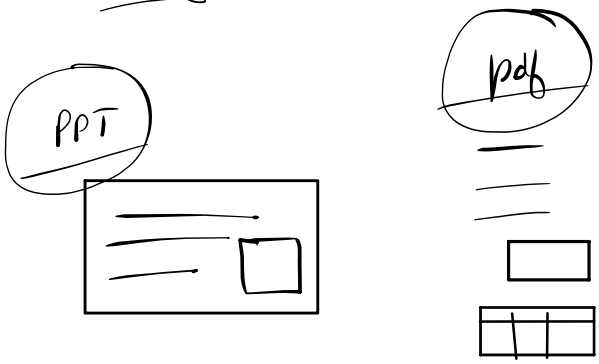
Documents → Text + images

1) Difficult to find
2) Preprocessing steps.

Mixed sources of input

images → text ✗

Text + images → Combined form



Mixed info

Mixed modalities

Pdf files → Text, Images, Tables

Base RAG

① Ingestion Phase

(Knowledge Source

↓

Knowledge Base → VS

↓

External Knowledge

② Retrieval Phase

Query → Retrieve (similar to the query)

① Ingestion Phase

knowledge source → Raw documents → Text

↓

file types. .pdf, .md,

① Ingestion Phase

① Document loaders → Load, Parse, Extract → Unified format

↓ file types. .pdf, .md, .txt, .py

Text ←

② Splitting → Chunks

Text encode.

③ Embedding vectors → Numerical Representation

Semantic meaning → Dense Representation

Dense Vectors

Doc / Text Embeddings.

Semantic meaning

dimensionality

features → values

nuances, patterns.

④ Vector stores → Postgres

Store the embeddings, Index them } Similarity search

Text Query → Query vector → VS → Similarity search

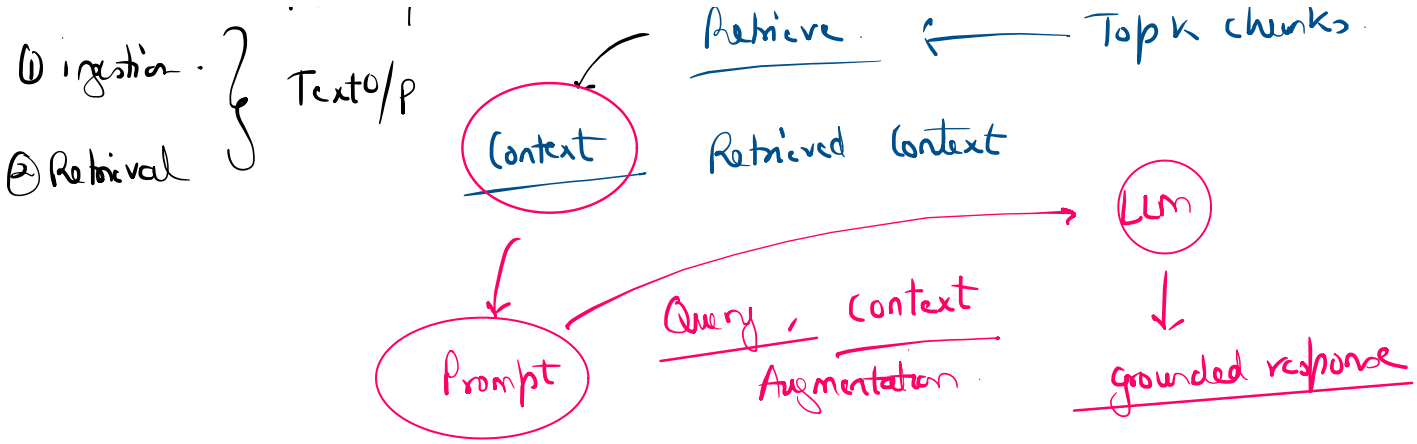
Input Prompt

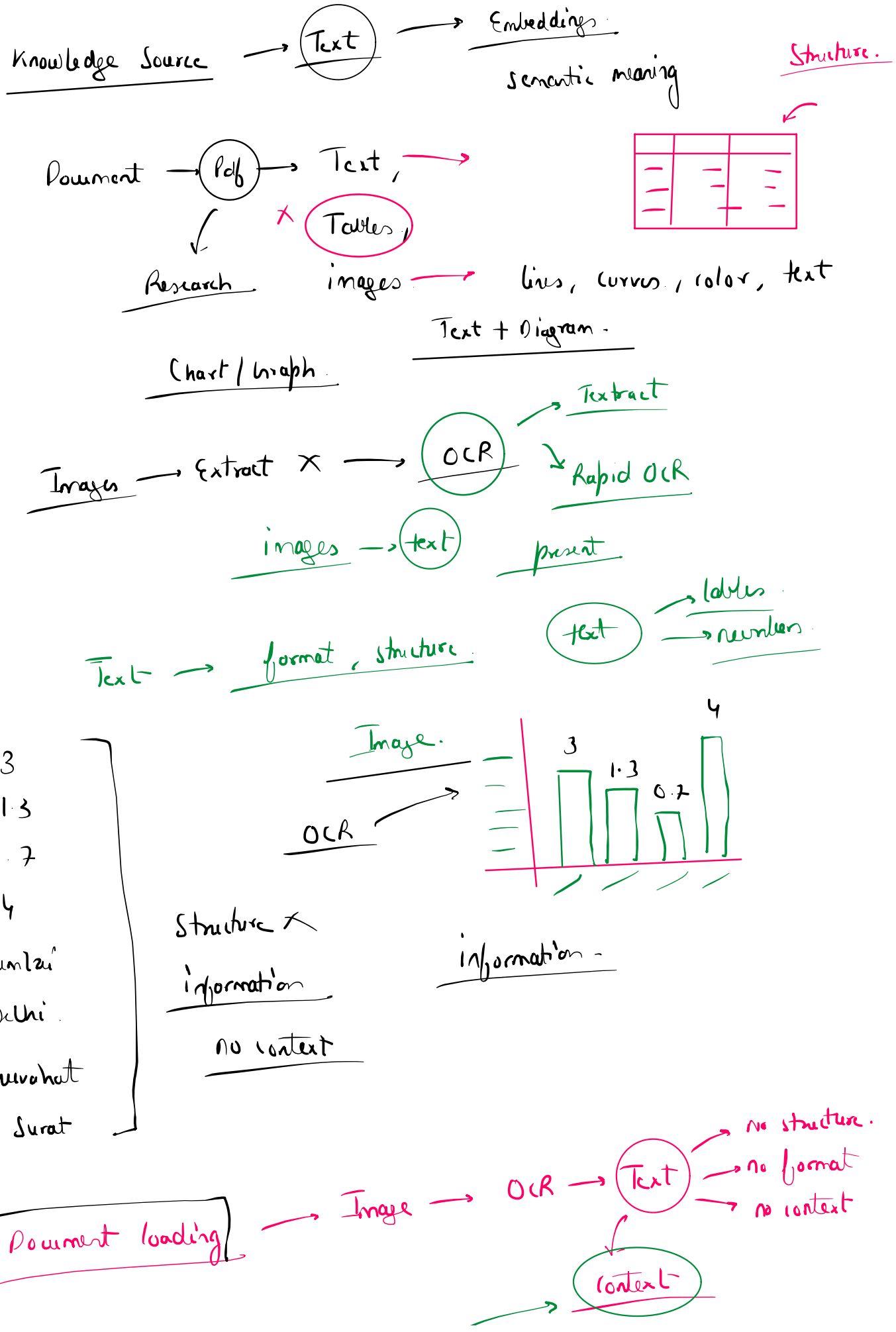
Embedding Vectors → Semantic similarity.

Text

Retrieve ← Top k chunks

① ingestion. Text I/P, Text O/P


1. ingestion
2. Retrival
Text o/p
Context
Retrieve
Top k chunks
Retrieved context
LLM
Prompt
Query + context
Augmentation
grounded response


Knowledge Source
Text
Embeddings
Semantic meaning
Structure.
Document
Pdf
Text,
Tables,
Research
images
lines, curves, color, text
Text + Diagram.
Chart / Graph
Images
Extract X
OCR
Textract
Rapid OCR
images
text
present
Text
format, structure
text
labels
numbers
3
1.3
0.7
4
Mumbai
Delhi
Guwahati
Surat
Image.
OCR
3
1.3
0.7
4
Structure X
information
no context
information.
Document loading
Image
OCR
Text
no structure.
no format
no context
context

confuse

embedding model → semantic meaning ✗ doc embedding ✗

Document → Text, Images → Text, Images

Unstructured → Text, Images, tables

Text chunks → [Document objects]

Images → Images extract [.png, .jpg]

Document → pdf

entity

metadata → { page no. — , section, source }

page_content → Text,

metadata → { source — , page no. }

embedding

embedding model

Text data

Doc → Text chunks · text → numbers

↓ Images

semantic meaning

↓

doc embeddings

Text → doc embeddings

Images → semantic meaning

Binary data.

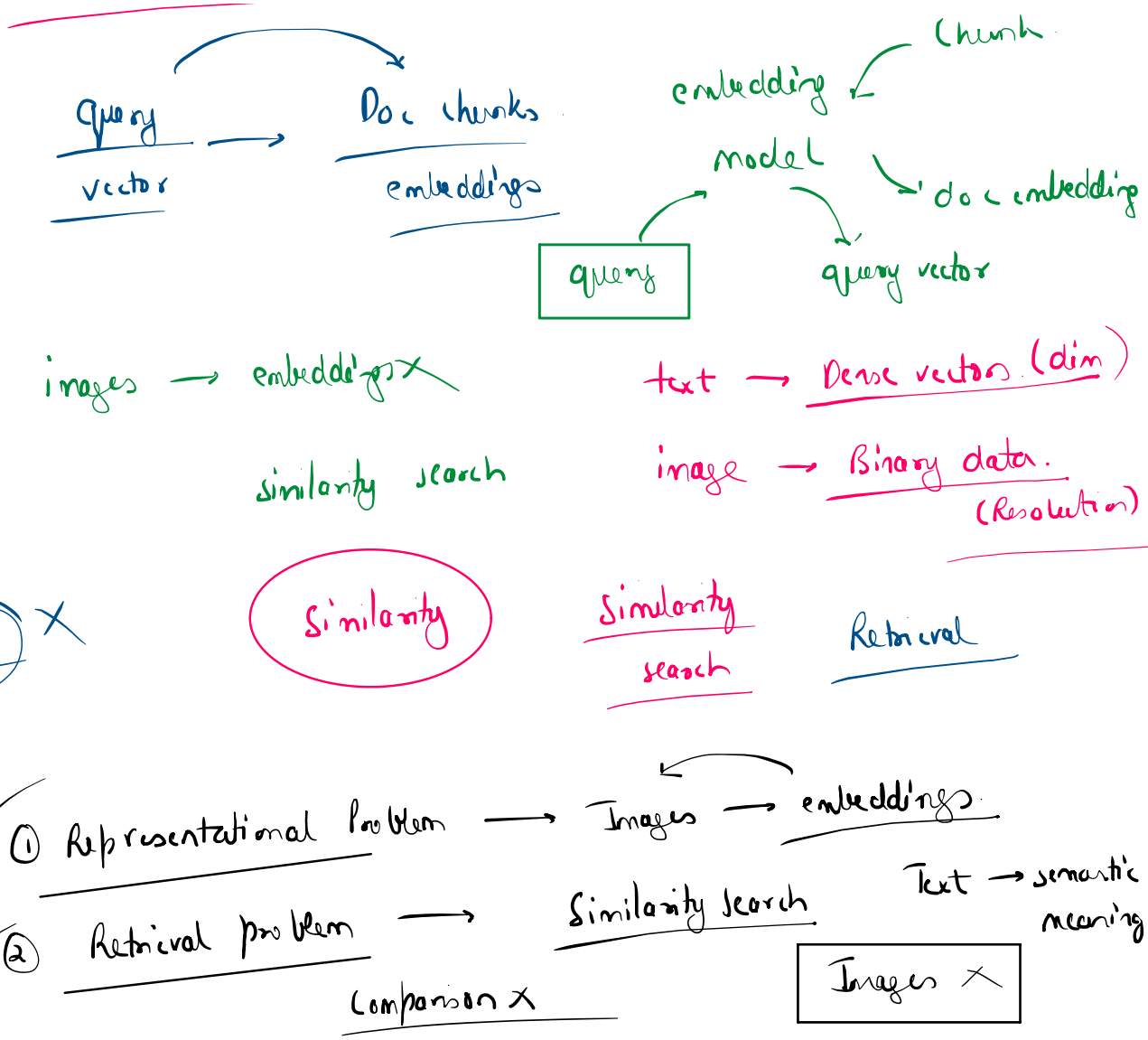
Images → embeddings ✗

text

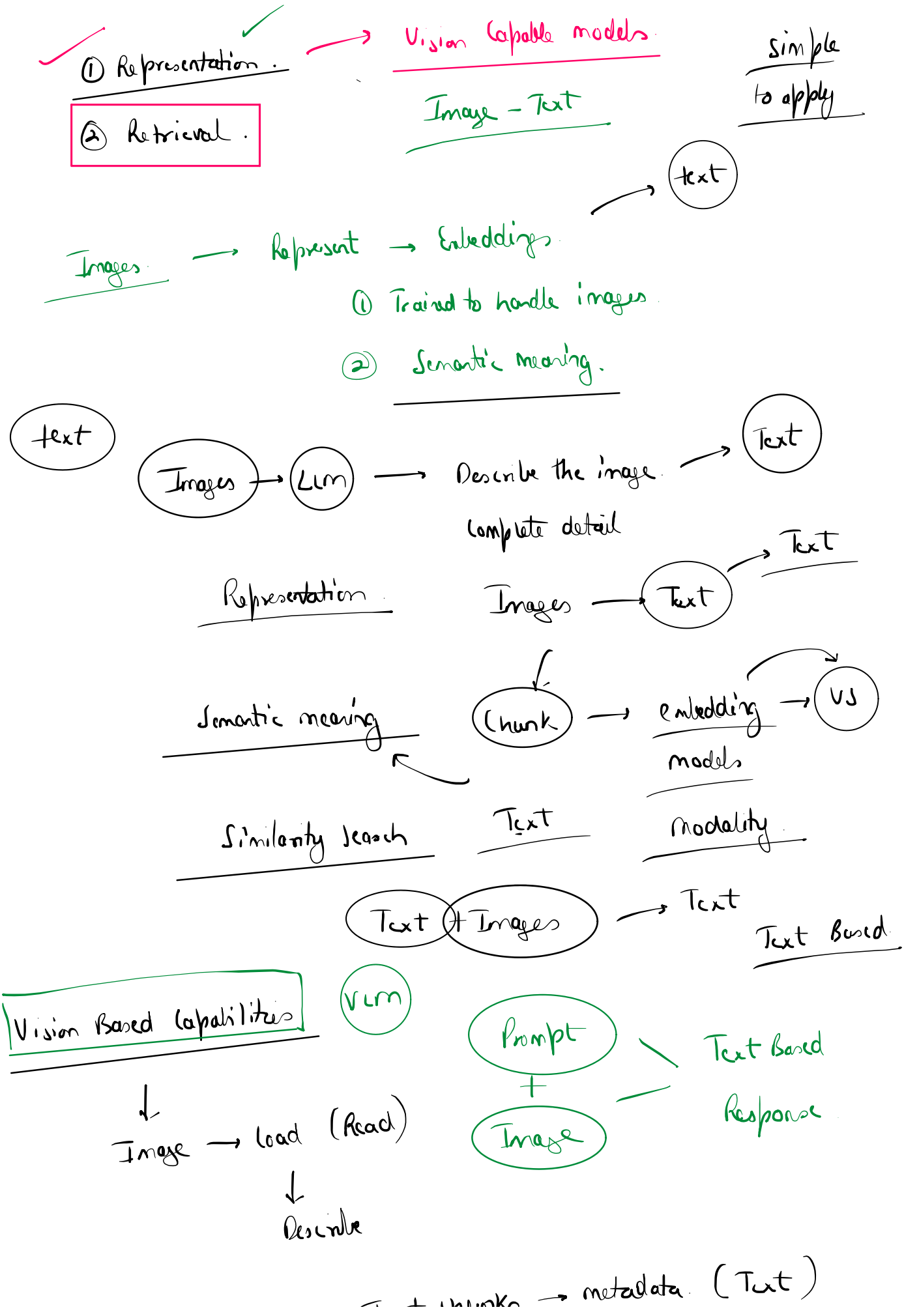
Collection of RGB pixel

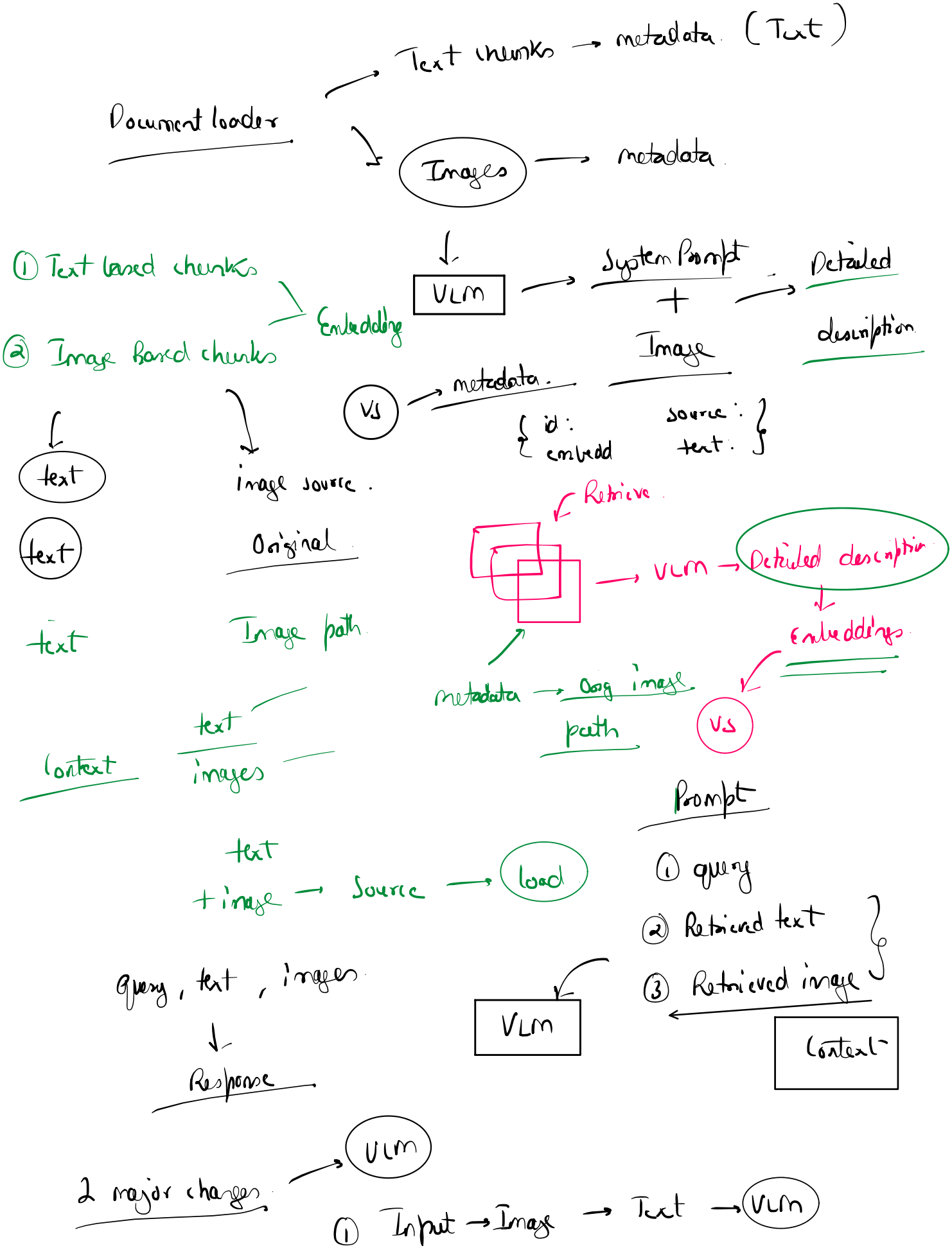
similarity search

Chunk


Query
vector
Doc chunks
embeddings
embedding
chunk
model
doc embedding
query
query vector
images → embeddings ✗
similarity search
text → Dense vectors (dim)
image → Binary data.
(Resolution)
RAG ✗
Similarity
Similarity search
Retrieval
① Representational problem → Images → embeddings
② Retrieval problem → Similarity search
Comparison ✗
Text → semantic meaning
Images ✗
Strategy 1 → More common (less accurate)
Strategy 2 → less common (More Accurate)

Strategy-1


① Representation.
② Retrieval.
Vision Capable models
Image - Text
Simple to apply
Images → Represent → Embeddings
text
① Trained to handle images
② Semantic meaning.
text
Images → LLM → Describe the image → Text
Complete detail
Representation
Images → Text → Text
Semantic meaning
Chunk → embedding → VJ
models
Similarity search
Text
modality
Text + Images → Text
Text Based
Vision Based Capabilities
VLM
Image → load (Read)
Describe
Prompt
+
Image
Text Based
Response
metadata (Text)


Document loader
Text chunks → metadata (Text)
Images → metadata
VLM
System Prompt + Image
Detailed description
① Text based chunks
② Image based chunks
Embedding
VS
metadata
{ id: embedd
source: text: }
text
text
text
image source.
Original
Image path
Retrive
VLM → Detailed description
Embeddings
metadata → Org image path
VS
context
text
images
text + image → Source → load
Prompt
① query
② Retrieved text
③ Retrieved image
Context
query, text, images
Response
VLM
2 major changes
① Input → Image → Text → VLM

② Generation → Text + Image → VLM

Bottlenecks

✗ ① VLM model → Accurate, Affordable ✗

embeddings.

Vision Transformers

100 images ↓ 100 detailed descriptions.

Image → detail describe. → hallucinate

Barchart → facts → Accurate

Text → hallucinate, miss out

② System Prompt ✗ → Description

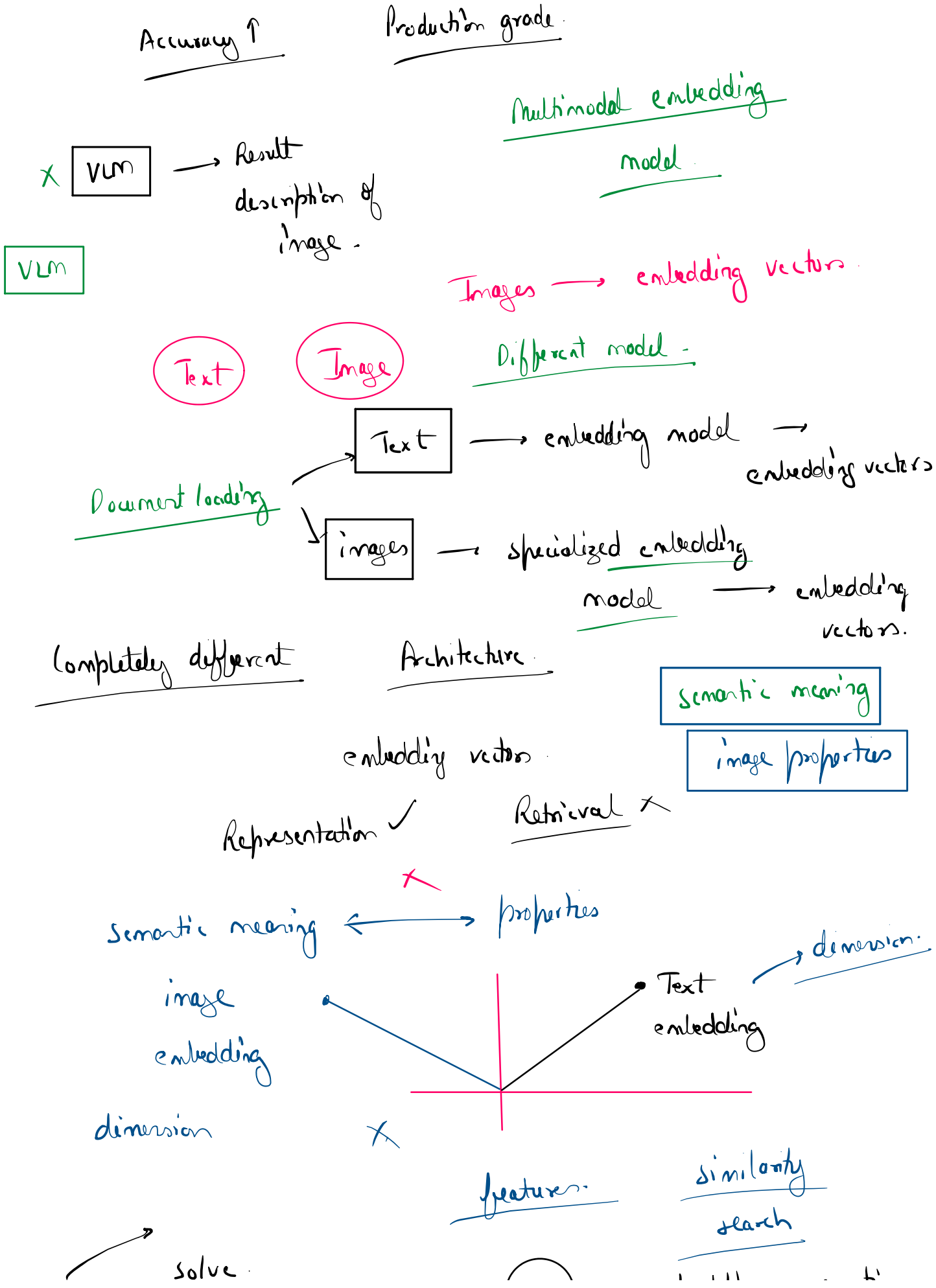
✓ Human error

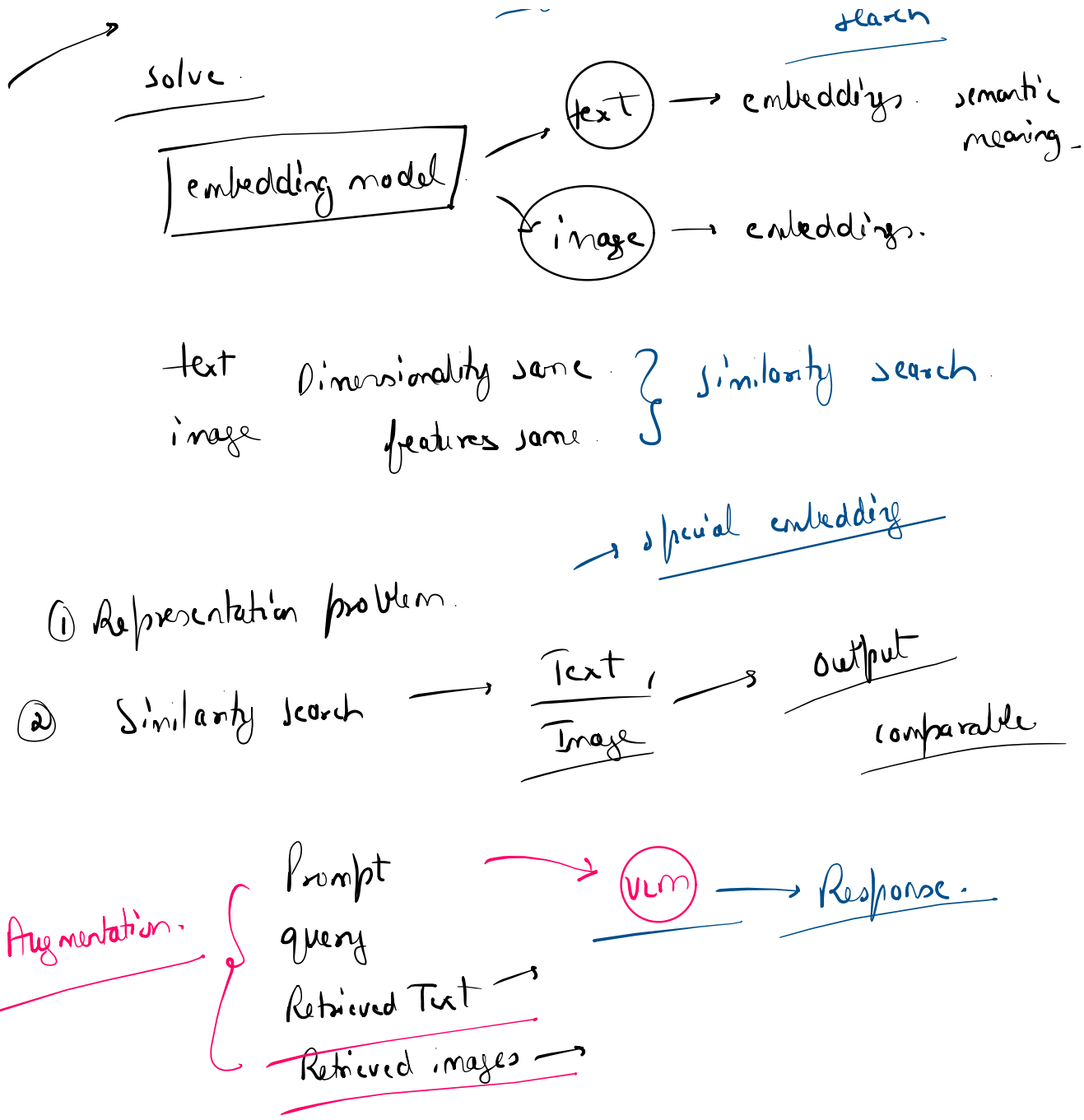
Instruction

weak VLM

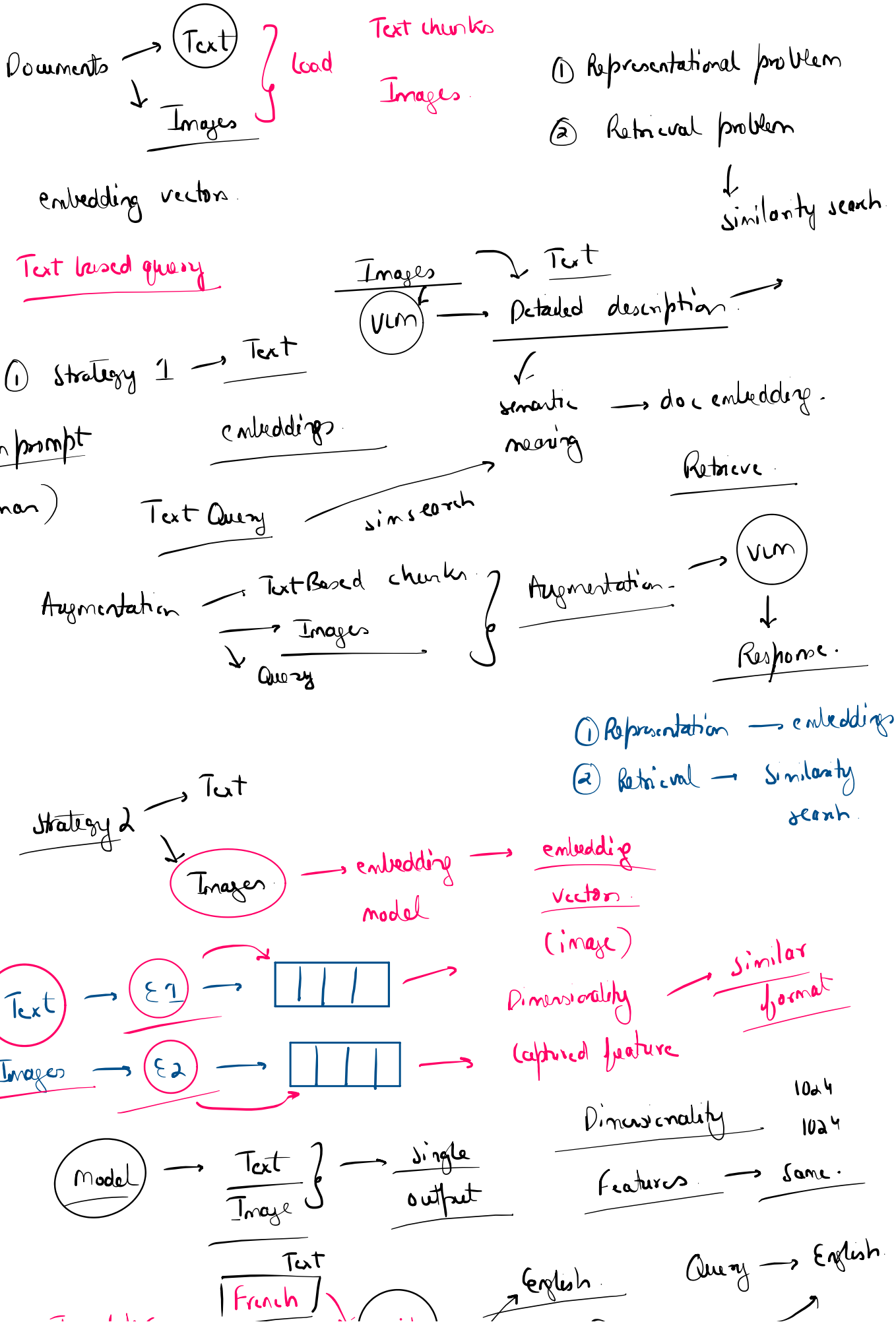
poor chunk

Strategy-2 (mostly used)




solve
search
embedding model
text
embeddings
semantic meaning
image
embeddings.
text
image
Dimensionality same
features same
similarity search
special embedding
① Representation problem
② Similarity search
Text / Image
output
comparable
VS
Augmentation.
Prompt
query
Retrieved Text
Retrieved images
VLM
Response.

CLIP(Contrastive Language–Image Pre-training)



Translator

Text
French
German
Image
English

English
Query
Unified language → Global language

CLIP model

Contrastive Language-Image Pre training Model.

LLM → Pre training
Massive dataset

Massive 400 m pairs

Contrastive → Loss function → Way to train the model

Text
Images

Images
Captions
Image
Text

Model

Language - Image

Relationship b/w
Image → Text
→ Single output

400 m Pairs

Image.
Short Description

Image: [photo of a golden retriever playing fetch]
Caption: "a golden retriever playing fetch in a park"

Image Enc →
Text Enc →

embeddings
loss minimize.
Text
embeddings.
similarity search

Retrieved text
Image
Path

Query

Similarity search

"a dog playing outside" → [0.21, -0.54, 0.87, ...]
[golden retriever image] → [0.23, -0.51, 0.85, ...] ← very close!

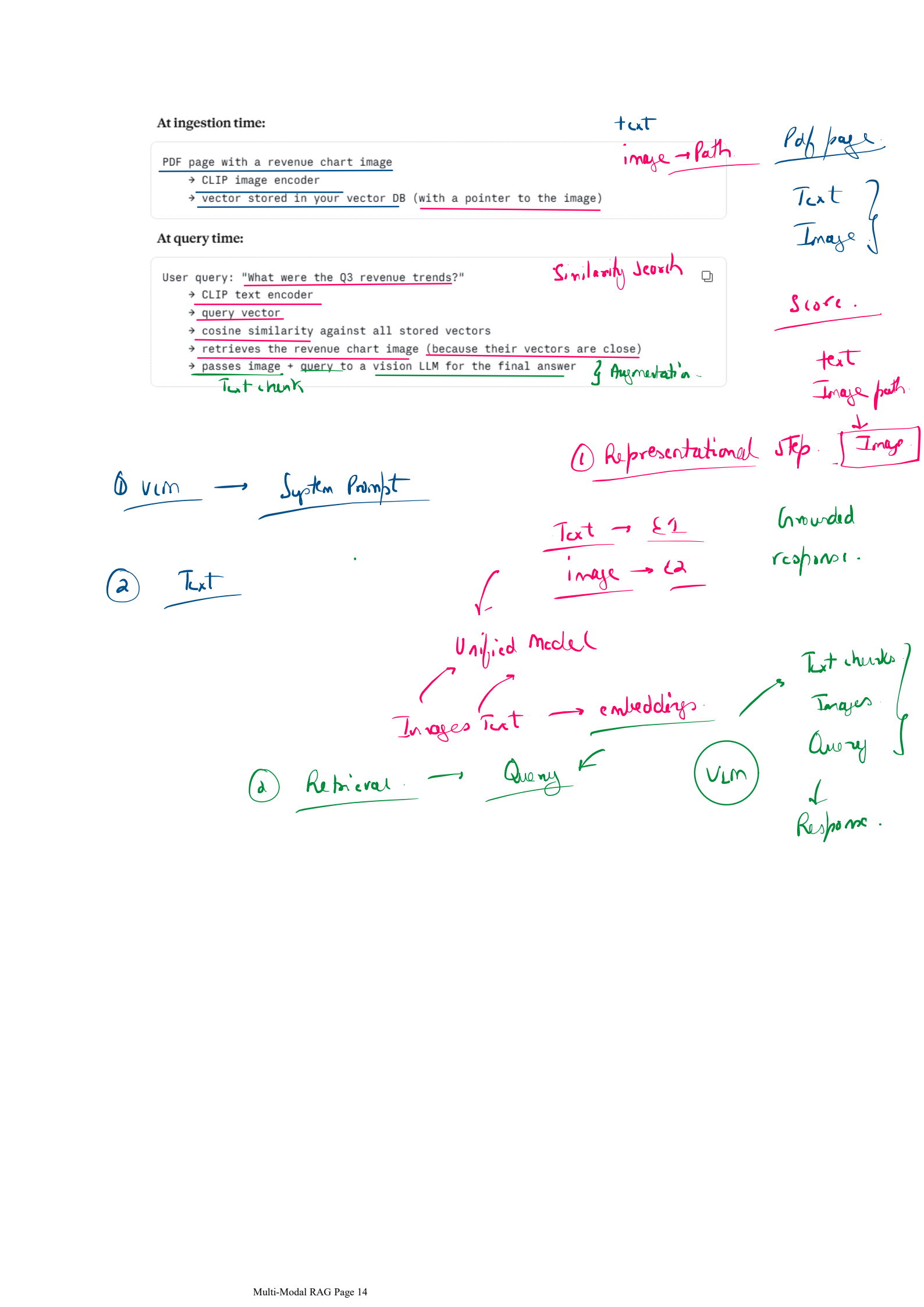
"quarterly revenue chart" → [0.91, 0.12, -0.33, ...] ← far from the dog vectors

At ingestion time:

```
PDF page with a revenue chart image
    → CLIP image encoder
    → vector stored in your vector DB (with a pointer to the image)
```

At query time:

```
User query: "What were the Q3 revenue trends?"
    → CLIP text encoder
    → query vector
    → cosine similarity against all stored vectors
    → retrieves the revenue chart image (because their vectors are close)
    → passes image + query to a vision LLM for the final answer
```



Embedding model → Text, Images → embeddings, same embedding space.

Text ↕ Relationship. Image

Query → Text Based. → Query vector → Similarity search → Dog Images embeddings + Query. Retrieve

Images + Text → Contrastive learning

Dataset → Image → embedding vector; Caption → embedding vector → Highly similar

2 aspects → Attractive → Image Dog → Caption Dog

→ Repulsive → Image Building → Caption Dog

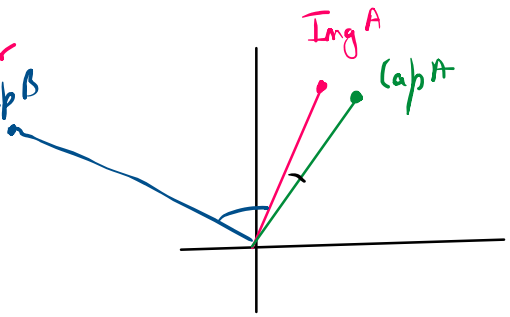
Accuracy.

20 18

A Img → A Cap Similar

B Img → A Cap Dissimilar

Contrastive learning



Contrastive learning

Image A	↔ Caption A	✅ (should be close)
Image B	↔ Caption B	✅ (should be close)
Image A	↔ Caption B	❌ (should be far apart)
Image B	↔ Caption A	❌ (should be far apart)

4

Image Caption pairs

Encoders.

```
Pair 1: [Image of a dog]       ↔ "a dog playing in the park"
Pair 2: [Image of a revenue chart] ↔ "Q3 revenue increased by 20%"
Pair 3: [Image of a mountain]  ↔ "a snowy mountain at sunset"
Pair 4: [Image of a cat]       ↔ "a cat sleeping on a couch"
```

Text Enc

Image enc

4 images

```
Image encoder output (raw):
  I1 = [2.1,  0.3, -1.4,  0.8]  ← dog image
  I2 = [0.5,  1.9,  0.2, -0.7]  ← chart image
  I3 = [-1.2, 0.6,  1.8,  0.4]  ← mountain image
```

← Image Vectors

```
Image encoder output (raw):
  I1 = [2.1,  0.3, -1.4,  0.8]   ← dog image
  I2 = [0.5,  1.9,  0.2, -0.7]   ← chart image
  I3 = [-1.2, 0.6,  1.8,  0.4]   ← mountain image
  I4 = [1.9,  0.1, -1.3,  0.7]   ← cat image

Text encoder output (raw):
  T1 = [1.8,  0.4, -1.2,  0.9]   ← "a dog playing..."
  T2 = [0.6,  1.7,  0.3, -0.8]   ← "Q3 revenue..."
  T3 = [-1.0, 0.5,  1.9,  0.3]   ← "a snowy mountain..."
  T4 = [1.7,  0.2, -1.1,  0.6]   ← "a cat sleeping..."
```

Image Vectors

Captions

N → Pair

1 Image Vector

4 caption vectors

Captions

Similarity matrix

Similarity

NxN

Cosine Similarity

16 sim.

	T1 (dog)	T2 (chart)	T3 (mtn)	T4 (cat)
I1 (dog)	0.98	0.05	0.02	0.81
I2 (chart)	0.06	0.97	0.08	0.04
I3 (mtn)	0.03	0.07	0.96	0.05
I4 (cat)	0.79	0.03	0.04	0.97

Images

Image A → cap A

Attractive.

Loss.

Repulsive.

{ Image A → cap A ; Image B → cap B }

Cheat

Image A [0, 0, 0, 0, 0]

cap A [0, 0, 0, 0, 0]

Similarity 1

Attractive loss

Image → cap.

Image B [0, 0, 0, 0, 0] 1

Cap B [0, 0, 0, 0, 0]

Training

Representational collapse

Image embeddings

Caption embeddings.

Repulsive.

Img A → cap A ; Sim ↑ Attractive.

Very close to each other

Representational collapse.

Img A → cap B } sim ↓ Repulsive.

As far as possible

Star

gravity

Massive.

Fusion.

Balance.

Balancing act

Img A → cap A

Maximise

Attractive

Similarity Matrix

	T1 (dog)	T2 (chart)	T3 (mtn)	T4 (cat)
I1 (dog)	0.98	0.05	0.02	0.81
I2 (chart)	0.06	0.97	0.08	0.04
I3 (mtn)	0.03	0.07	0.96	0.05
I4 (cat)	0.79	0.03	0.04	0.97

Loss

Contrastive.

Categorical crossentropy → ML, DL } Multi class classification

-1

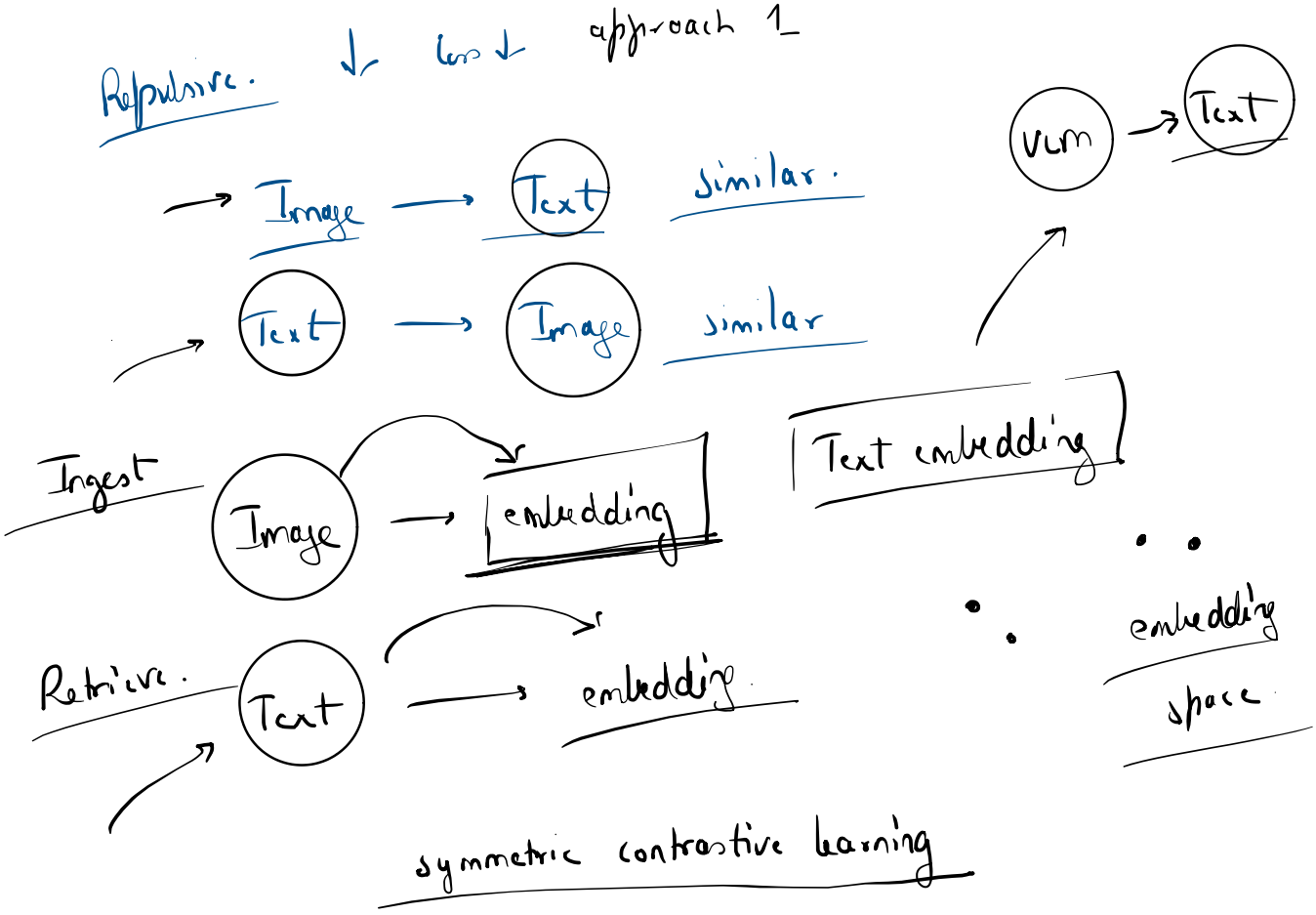
1

actual class → Score. → 0

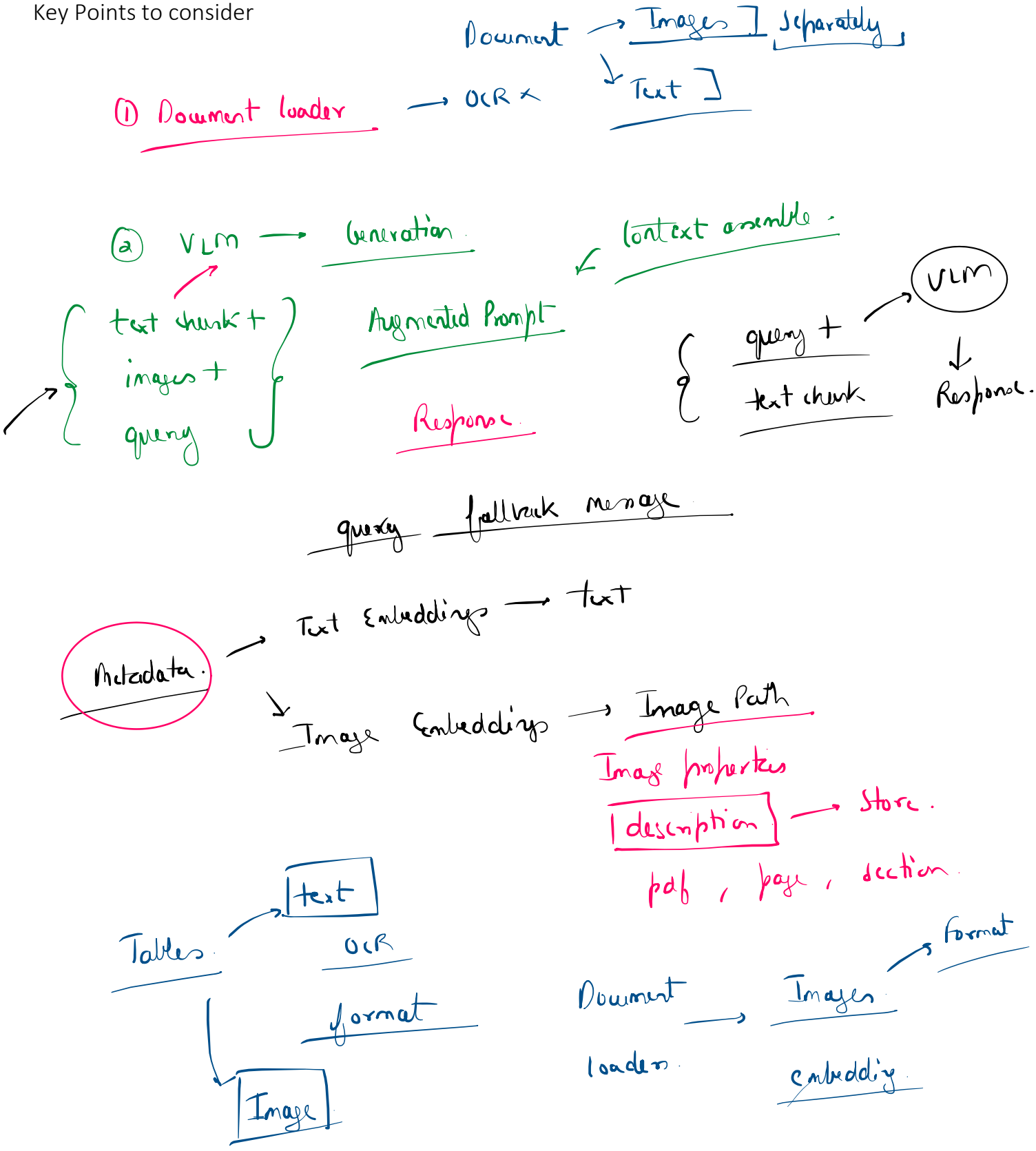
Scores in the diagonals should be approaching 1 → Attractive.

loss ↓

↑ Repulsive Scores at non-diagonal places should not


approach 1
Repulsive.
Image
Text
similar.
Text
Image
similar
VLM
Text
Ingest
Image
embedding
Text embedding
Retrive.
Text
embedding
embedding space
symmetric contrastive learning

Key Points to consider



Code